ASCN

ASEAN SMART CITIES NETWORK

Monitoring & Evaluation Report 2023

one vision
one identity
one community

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List of ASCN Projects across the Six Focus Areas **20**

List of Acronyms

AASCTF	ASEAN Australia Smart Cities Trust Fund
ADB	Asian Development Bank
ADMS	Advanced Distribution Management System
CBA	City Bus Authority of Phnom Penh
CRM	Customer Relation Management
DBKL	*Dewan Bandaraya Kuala Lumpur* (Kuala Lumpur City Hall)
DICT	Department of Information and Communications Technology of the Philippines
DEPA	Digital Economy Promotion Agency of Thailand
DPTMP	Davao Public Transport Modernisation Project
ESS	Energy Solar System
HPBIS	High Priority Bus Information System
GIS	Geographic Information System
IED	Intelligent Electronic Device
IOC	Integrated/Intelligent Operation Centre
IRDA	Iskandar Regional Development Authority
JICA	Japan International Cooperation Agency
JTEC	Japan Telecommunications Engineering Consultancy
KETI	Korea Environmental and Industrial Technology Institute
KLUO	Kuala Lumpur Urban Observatory
KKCH	Kota Kinabalu City Hall
NPTDC	Nay Pyi Taw Development Committee
MIC	Ministry of Information and Communications of Viet Nam
MCDC	Mandalay City Development Committee
MLIT	Ministry of Land, Infrastructure, Transport, and Tourism of Japan
MoU	Memorandum of Understanding
PIBO	Project for Improvement Bus Operation
RMMT	River Management and Monitoring Tool
SCADA	Supervision Control and Data Acquisition
SCAP	Smart City Action Plan
SDP	Social Development Program
SRPA	Siem Reap Provincial Administration
SMA	Sarawak Multimedia Authority
SmartJAMP	Smart City Supported by Japan ASEAN Mutual Partnership
SME	Smart and Medium-sized Enterprises
TOD	Transit Oriented Development
USTDA	United States Trade and Development Agency
YCDC	Yangon City Development Committee

1| Introduction

The ASEAN Smart Cities Network (ASCN) continues to strengthen efforts in advancing smart city development across the region through the respective smart city projects as well as spearheading the regional cooperation initiatives.

Exhibit 1 presents a snapshot of the share of ASCN projects in each focus area. These projects have expanded from the initial list indicated by the respective ASCN Cities in the ASCN Smart City Action Plans (SCAPs) during the establishment of ASCN in 2018. The ASCN projects have expanded both in numbers as well as in scope of activities within the projects. As of August 2023, the number of smart city projects under the ASCN stood at 86.

At the regional level, the ASCN has commenced the project to develop the ASEAN Smart City Investment Toolkit this year. The Toolkit will provide the ASCN and other ASEAN cities with knowledge and understanding of various options available for funding and financing smart city initiatives.

From February-June 2023, the ASCN also convened a series of discussions on various topics under the six focus areas. The ASCN Discussion Series served as a useful avenue for ASCN and other ASEAN Cities to learn from each other's experience and exchange views with experts in solving various urban challenges.

Exhibit 1: ASCN Projects across the Six Focus Areas as of August 2023

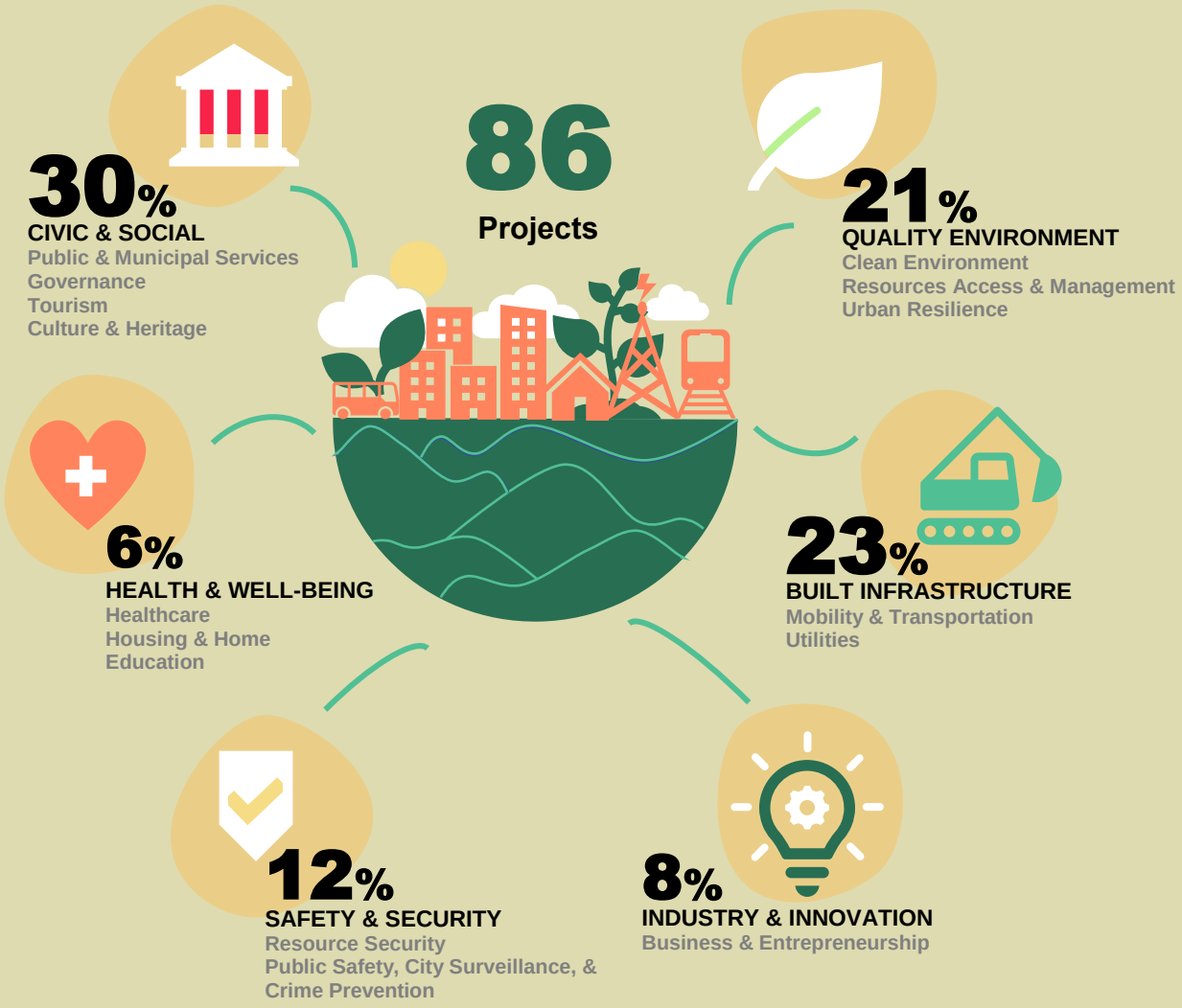


Exhibit 2: Overall Implementation Progress of ASCN Projects as of August 2023

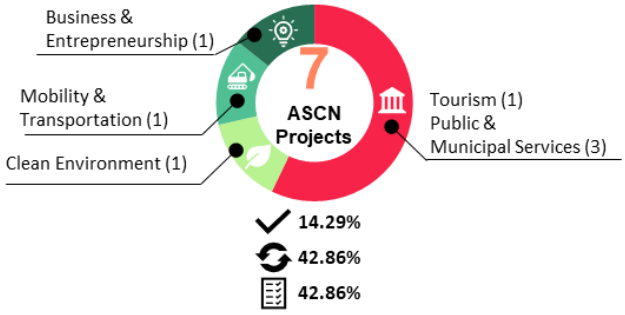
The overall implementation progress of ASCN projects can be seen from **Exhibit 2**, where around 77.9% of the projects are on-going with 19.8% in planning stage, and 2.3% completed. Depending on the nature of the projects, the on-going projects are typically those beyond feasibility stage. The on-going projects may also include efforts to advance or scale-up the existing works/outputs.

Country Chart in **Exhibit 3** depicts the number of ASCN projects by country, distribution of ASCN projects by focus areas and specific types of projects.

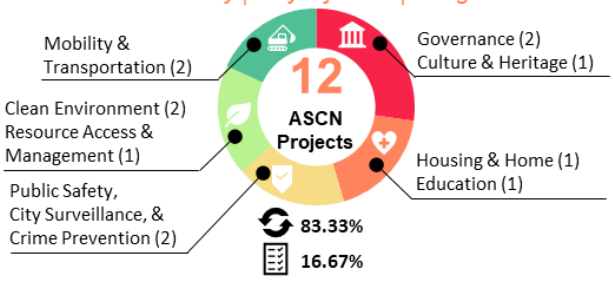
The ASCN M&E Report 2023 presents the status of ASCN projects by focus area with each ASCN city having one of their projects highlighted. Each ASCN city has one of their projects highlighted. The full list of ASCN projects status can be found in the Appendix. The Report also provides a summary of observations and recommendations.

Exhibit 3: Country Chart as of August 2023

BRUNEI DARUSSALAM Bandar Seri Begawan



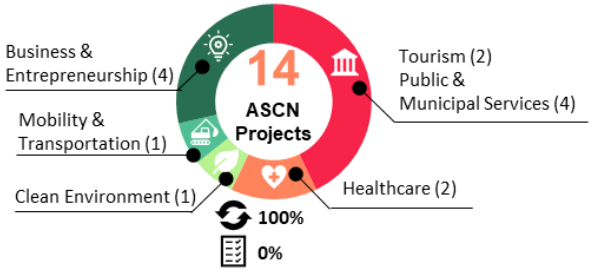
MYANMAR Mandalay | Nay Pyi Taw | Yangon



CAMBODIA Battambang | Phnom Penh | Siem Reap

THE PHILIPPINES Cebu City | Davao City | Manila

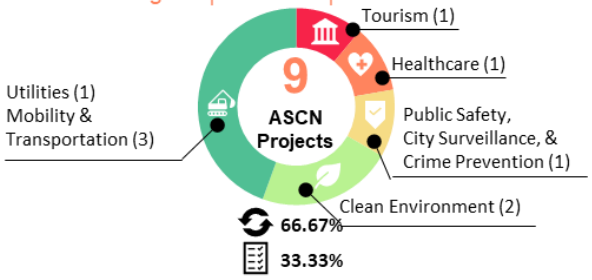
INDONESIA Banyuwangi | Jakarta | Makassar



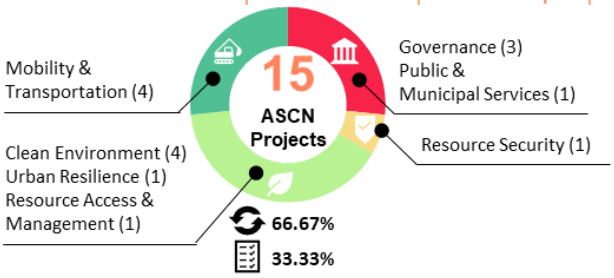
SINGAPORE Singapore

LAO PDR Luang Prabang | Vientiane

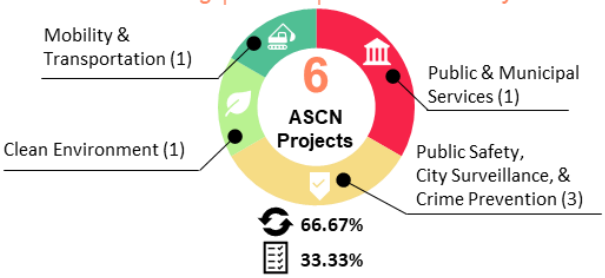
THAILAND Bangkok | Chonburi | Phuket



MALAYSIA Johor Bahru | Kota Kinabalu | Kuala Lumpur | Kuching



VIET NAM Da Nang | Ha Noi | Ho Chi Minh City



Civic & Social
Health & Well-Being
Safety & Security
Quality Environment
Built Infrastructure
Industry & Innovation
Completed
On-going
Planning

2| Status of ASCN SCAPs

Civic & Social

The focus area of Civic and Social aims to achieve social cohesion, promote social equity, preserve and deepen appreciation for cities' cultural authenticity and heritage, promote the tourism sector, and improve the citizen experience. These involve enhancing good governance in the delivery of public and municipal services, decision making, transparency, accountability, and information accessibility. Civic and Social accounts for 30% of the overall ASCN projects. Siem Reap, Kuala Lumpur, Mandalay, Yangon, Phuket, and Ho Chi Minh City are examples of ASCN Cities that have projects in this focus area.

Siem Reap, Cambodia

Following the finalisation of Siem Reap Smart City Roadmap, the Siem Reap Provincial Administration (SRPA) has continued the effort to implement the pilot projects across four sectors under the Roadmap, towards realising an international tourist destination that is convenient, safe, healthy, and sanitary for memorable tourism experiences.

Smart Tourist Management System

Objective: To be a liveable, smart, clean, safe and sustainable city for both local residents and tourists alike, through the use of advanced technologies.

The Roadmap identifies six pilot projects, namely: (1) illegal parking monitoring; (2) waste collection improvement; (3) tourism statistics digitalisation; (4) government internal document tracking; (5) urban environment monitoring by sensors; and (6) public-private tourism collaboration. The pilot projects would utilise and manage public data, citizen data, area data, private sector data, and individual data to update tourism services and meet tourists' needs.

Further to the support provided for the assessment survey during the Roadmap formulation, Japan International Cooperation Agency (JICA) has continued to provide technical assistance until 2025 to implement the pilot projects. A Memorandum of Understanding (MoU) has been signed between Siem Reap Tourism Club and Nippon Koei Co. Ltd., to promote sustainable and smart tourism in Siem Reap through establishing a digital tourism platform. Both parties as well as SRPA and Ministry of Tourism will collaborate to establish the Cambodia Tourism Promotion Marketing Board (Siem Reap Branch) through public private partnership.

Kuala Lumpur, Malaysia

Kuala Lumpur City Hall/ *Dewan Bandaraya Kuala Lumpur (*DBKL) has developed G-Asset as a one-stop platform for geospatial data with a set of Geographical Information System (GIS) data that can be obtained, shared and updated online. Five modules had been fully developed and uploaded on G-Asset, namely: (1) Asset and Inventory Management; (2) Various Permits; (3) Contract Profile; (4) Mobile Data Updating; and (5) E-Complaint KLCares – mobile application.

Ways forward for the G-Asset include adding features to improve data collection for analytics and decision-making purposes, making G-Asset accessible to public as a single reference for spatial data, and expanding G-Asset towards establishing a Kuala Lumpur Urban Observatory (KLUO), a bigger platform for managing and disseminating urban data. Data ownership and updating, internet connectivity, and data usership are some challenges faced by DBKL in implementing this initiative.

Exhibit 4: Kuala Lumpur Urban Observatory/G-Asset

Objective: To develop a central data repository to collate, update and disseminate social, economic and physical data for planning purposes towards a more efficient city management.

Mandalay, Myanmar

To improve land and tax management system, the Mandalay City Development Committee (MCDC) has conducted the ground survey, field identification survey, and field verification survey for creating a GIS database as well as a land use map and topographic map for the Mandalay City. During 2020-2022, MCDC had digitised nearly 1,300 cadastral maps, covering 165,000 lots, within approximately 3,000 blocks.

Cadastral Map and GIS Database

Objective: To develop a GIS database, land use map, and topographic map for Mandalay City to implement the house numbering system for the whole city, improve land and tax management system.

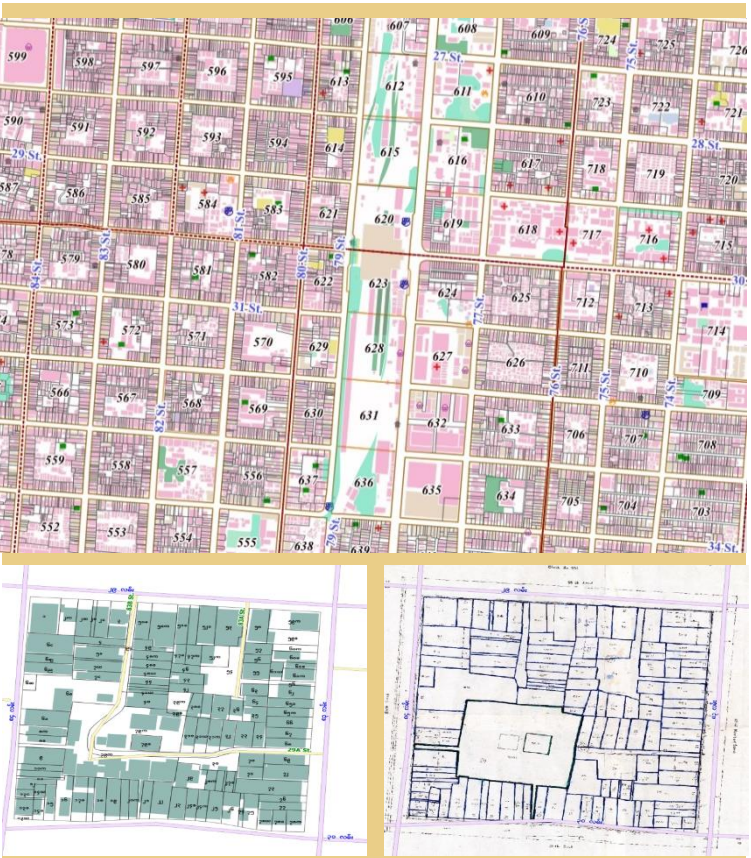


Exhibit 5: Transforming Hand Drawing into Digitised Cadastral Map for Each Block © MCDC

Based on the database, establishment of a house numbering system for the whole city will be proposed. Capacity building and technical assistance are needed to establish standards for an effective, efficient and consistent house numbering system in the Mandalay City. Moreover, digital infrastructure will be required for data processing, managing and linking the house numbering system with taxation and business license, and others.

Yangon, Myanmar

Linking people with the heritage that surrounds them has been the focus of Yangon's smart city development. Yangon City Development Committee (YCDC) has completed the improvement in roads and pedestrian walkway in the downtown area, as well as the sidewalk of Kandawgyi Lake, the lung of Yangon, near the downtown area. The lake sidewalk is 7307-feet long and around 18-feet wide. Since COVID-19, people have increasingly been relying more on bikes than public transport. Therefore, YCDC has built and operationalised a 120-km bike lane.

Conservation of Downtown Yangon

Objective: To boost Yangon's attractiveness as a tourist destination by improving the streetscapes, repurposing the heritage buildings, and defining the zoning regulations.

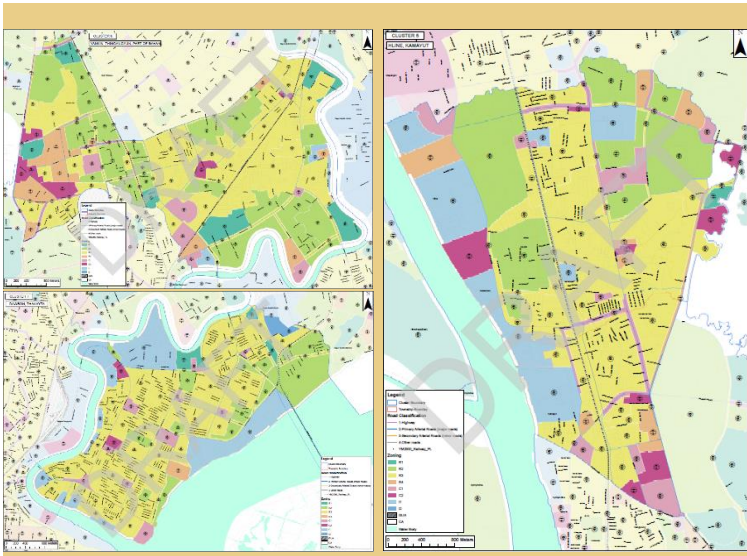


Exhibit 6: Pages from the Draft Yangon Zoning Map © YCDC

Furthermore, YCDC has been working on setting up a Zoning Regulation and Zoning Map to: (a) bring about sustainable development to Yangon while preserving its distinct characters and enhancing quality of life; (b) formulate integrated and uniform zoning regulation to achieve sustainable development of Yangon; and (c) achieve transparency in guiding and controlling the development of Yangon. The draft versions of the Zoning Regulation and Zoning Map can be accessed on www.ycdc.gov.mm.

Phuket, Thailand

The Digital Economy Promotion Agency (DEPA), Tourism Council of Thailand, and Big Data Institute have been working together for the TravelLink project to create Customer Data Platform for tourism industry.

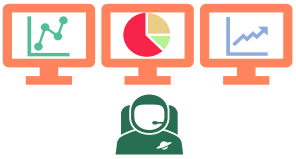
Utilising real-time data from various sources, TravelLink provides and consolidates more than 80 dashboards, including analytics on social media platforms, tourist behaviour, relevant trends and news. TravelLink also covers more than 1,400 tourist attractions.

Challenge remains in collecting tourism data from many sources and using data analytic and big data to analyse and visualise for benefit of the tourism industry. In this undertaking, DEPA has also partnered with Tourism Authority of Thailand, Tourism Council of Thailand, and Phuket Tourist Association.

Exhibit 7: City Data Platform
Objective: To build big data that allows for better understanding of residents and tourists in Phuket, supporting the city administration with more efficient and accurate planning of infrastructure, public utilities, common amenities, as well as, supporting the private sector in their analysis of the tourism industry.

Ho Chi Minh City, Viet Nam

Ho Chi Minh City has implemented a pilot version of the Integrated Operation Centre based on camera integration and application of some video analytic functions. The city has continued the preparation to develop a full-functioning Integrated Operation Centre with technical assistance from United States Trade and Development Agency (USTDA).



Integrated Operation Centre
Objective: To integrate daily information and resources across all sectors in Ho Chi Minh City; collect, integrate, and analyse data and information from CCTV, sensors, and operation centres in all government sectors; and serve as the 'brain' of the Smart City.

In the meantime, the development of a portal has commenced for providing information and responding to public requests. The portal is envisaged to allow citizens and enterprises to send their feedback, reports on public infrastructure incidents and public transportation systems, towards improving public services and public administration.

Health & Well-Being

Improving the quality and access to healthcare, housing, and education in a way that is innovative and resource-efficient are the core of Health and Well-Being focus area. Health and Well-Being accounts for 6% of the overall ASCN projects. Examples of ASCN projects fall under this focus area can be found in Nay Pyi Taw and Makassar.

Exhibit 8: Affordable and Low-Cost Housing
Objective: To construct affordable and low-cost housing for the public and government employees, including the retired government employees.

Nay Pyi Taw, Myanmar

Ensuring health and well-being of citizens contributes to the improvement of living standards. Nay Pyi Taw Development Committee (NPTDC), Nay Pyi Taw Council, the Ministry of Construction have joined forces to deliver affordable and low-cost housing for the public and government employees, including the retired government employees. As of April 2023, the effort has resulted in 90% completion of the entire 3,000 units of housing, including the basic infrastructures. The public, including retired government employees, have occupied the completed units. Continued challenges are presented by high cost of construction materials and low land prices in Nay Pyi Taw which require NPTDC to consider sustainability aspect of the project. The culture of owning land have partly influenced the uptake of affordable and low-cost housing.

Mobilising resources through the collaboration with the private sector and banking sector is necessary to ensure that homes can enjoy easy access to the commercial complex and public amenities within the project area.

Makassar, Indonesia

Makassar City Government through its Smart Healthcare programme called *Dottoro'ta* or "our doctor" in Makassar language, continues to provide homecare and telemedicine services for the citizens. In 2022, *Dottoro'ta* attended to more than 4,700 patients.

To accelerate the implementation of the homecare and telemedicine services, the following efforts have taken place, namely, Memorandum of Understandings (MoUs) signed between the Public Health Office (*Puskesmas*) of Makassar City Government with Makassar University Hospital and Association of Indonesian Cardiovascular Specialists (Makassar Chapter) respectively; electrocardiography (ECG), ultrasonography (USG), and tele-spirometry trainings for *Puskesmas* doctors; homecare service trainings for *Puskesmas* doctors and nurses; as well as trainings on the flow of homecare service for *Puskesmas* doctors, nurses, administrative staff, and drivers. The development of the mobile-app version of *Dottoro'ta* is on-going

Challenges that *Dottoro'ta* has encountered include high cost in procuring and maintaining the infrastructure, current focus on clinical service, high operating cost for telemedicine and home visits, conventional mindset, i.e. doctor and patient need to be in the same room for medical care which has influenced the outreach of *Dottoro'ta*.

Exhibit 9: Smart Health Care
Objective: To develop a healthcare ecosystem that ensure that all citizens have easy and direct access to comprehensive healthcare services towards improving health independence and reducing impacts of diseases.

47 Puskesmas
48 Homecare cars
46 Tele-USG
46 Tele-ECG
7 Tele-spirometry

Safety & Security

The focus area of Safety and Security refers to the adoption of effective technologies to address urban security problems, such as, strengthening cybersecurity of networked infrastructure and objects, as well as public safety, city surveillance and crime prevention, among others.

Safety and Security accounts for 12% of the overall ASCN projects. Cebu City, Manila, Da Nang, and Ha Noi are examples of ASCN Cities that have projects in this focus area.

Cebu City, the Philippines

Cebu City has been developing the Automated Citywide Traffic Control System through which it intends to upgrade and modernise the traffic infrastructure which can also effectively synchronise traffic flow and movements in key intersections within the city. Phase 1 installation covering 18 intersections in the uptown and midtown areas has been completed which includes the installation of automated traffic-controlled system which adapts the timings of traffic lights based on real-time traffic conditions to optimise the flow of traffic. Inter-agency coordination is key to ensure full operation of the installed infrastructure.

Phase 2 installation covering 27 intersections in the downtown areas is at 80% completion. The underground pipes and wirings of utility companies need to be aligned with the CCTV cabling to facilitate the completion of Phase 2. Cebu City has also constructed a command centre to fully operate the control system.

Manila, the Philippines

Manila City has continued to improve its command centre to better respond to the security needs of its residents and visitors. Located at the Manila City Hall, the Command Centre serves as the main contact centre and dispatch for emergency services. Supported with high-definition wireless cameras installed all over the city and high-definition panels in place, the Command Centre provides real-time monitoring of city streets for possible flooding and accidents. Hotlines are available 24/7 and manned with agents and first responder units. This enables for the City's disaster and emergency team to provide better and faster response.

Additional funding support and technology upgrade for continuous system enhancements and iteration are needed to ensure accurate generation of data. The Command Center is also in need of IT experts and manpower to ensure that all emergency calls and reports are being responded to in the most efficient and fastest way.

© Manila City

© Cebu City

Exhibit 10

Cebu City Automated Citywide Traffic Control System
Objective:
To upgrade and modernise the traffic infrastructure which can also effectively synchronise traffic flow and movements in the key intersections within the city.

Manila City Command Centre Upgrade
Objective:
To rehabilitate and re-orientate the existing Command Centre to be more responsive to concerns of the residents and visitors.

Da Nang
Intelligent Operation Centre
Objective:
To build a centre to collect urban information and data and display it in real time on big screen; store and analyse to support timely decision-making; coordinate all forces to handle and respond issues and incidents in the city, especially in emergency situations.

Ha Noi
Intelligent Operation Centre
Objective:
To develop a centralised management system for monitoring and controlling signals and sensors, and for capturing visual information and resources across all sectors to effectively coordinate transport management, help in decision making, and respond to emergencies.

Da Nang, Viet Nam

Da Nang has been developing detailed construction design to construct an integrated Intelligent Operation Centre (IOC) consolidating the existing IOC with other existing specialised monitoring centres, namely the Traffic Monitoring Centre and Security Command Centre. The development of an integrated IOC project will include the construction of a city-wide IOC and several district operation centres, including establishing the Monitoring Room and Situation Room, management and monitoring system, which are connected to specialised systems and cameras for analysing and supporting decision-making.

A current IOC, or mini IOC, has been running at the Da Nang Public Service Information Centre. It gathers data from the existing services to serve pilot monitoring according to the guidance of the Ministry of Information and Communications (MIC). So far, basic smart city services have been deployed, namely: (1) complaint service; (2) traffic monitoring service; (3) order security monitoring service; (4) social network monitoring service; (5) information security monitoring service; and (6) public service monitoring.

The development of the integrated IOC requires the provision of skilled human resources and legal frameworks, including standards, codes, and investment model.

Ha Noi, Viet Nam

Ha Noi has been working on developing an IOC to contribute towards achieving its vision of becoming a green, culturally-rich, civil and modern city with sustainable development to create a better life for the people.

The IOC is envisaged to consist of Information Security and Safety Monitoring Centre; Support Centre for City's IT staff; Centre for Management of Press Information and Media Information; Center for Supervision, Traffic Management and Crime Prevention in Public Places; Data Analytics Centre; Question and Answer Centre; City Public Service Management Centre; as well as Centre for Receiving and Processing Information on Emergency, Fire Prevention, Search and Rescue, Medical Rescue.

To support the implementation of this project, Ha Noi has sought to address the limited know-how and technology, lack of skilled human resources, and lack of legal frameworks, including standards, codes, and investment model.

Quality Environment

The focus area of Quality Environment aims to leverage technologies to, for example, maintain a clean and pleasant environment; promote the sustainable use of ecosystems, natural resources and biodiversity; and strengthen resilience against disaster risks and potential climate change impact. Quality Environment accounts for 21% of the overall ASCN projects. Battambang, Luang Prabang, Vientiane, Johor Bahru, and Kuching are examples of ASCN Cities that have projects in this focus area.

Battambang, Cambodia

Battambang has intensified efforts to manage solid and liquid waste. The city has completed the construction of a new 20.8-hectare landfill and works are underway to expand the drainage and sewage system as well as to construct two new wastewater treatment plants. The Ministry of Public Work and Transport and Asian Development Bank (ADB) have been supporting Battambang in this undertaking.

Solid and Liquid Waste Management
Objective: To establish sewage and wastewater management systems, including developing additional drain and sewage infrastructure.

Battambang is setting up a mechanism to operate the new landfill, with selected staffers are being trained and prepared to start operating the landfill soon. To complement the current work, the city will develop a smart waste collection system (integrating GPS, GIS, and online payment), expand the catchment area of drainage and sewage networks, rehabilitate the main canal, and develop a faecal sludge treatment plant. The city is exploring technical and financial assistances to support these undertakings.

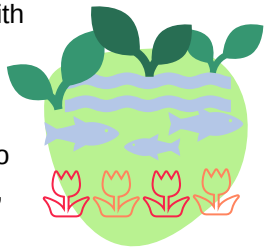
Battambang has also completed its Master Planning Study for Smart City Development supported by MLIT Japan, through the Smart City supported by Japan ASEAN Mutual Partnership (SmartJAMP). The Masterplan serves as Battambang's long-term vision for smart city development as well as a guideline for the use of smart approaches as Battambang develops and updates its land use plan and various infrastructure development plans in the future.

Among the priority activities suggested in the Masterplan is developing waste management system to promote waste segregation and recycling. The system is envisaged to utilise sensors and install communication equipment in the trash bins throughout the city to monitor and send the information to a control centre, which will analyse efficient collection routes and transmit the information to packer trucks to ensure efficient collection.

Luang Prabang, Lao PDR

Wetlands restoration and management continues to be a priority for Luang Prabang in preserving its heritage.

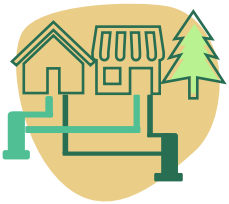
Today, six ponds in the World Heritage Site have been restored with funding and technical support from JICA. The restoration includes cleaning, stone installation to prevent landslide, wastewater treatment and drainage construction, fence installation protect ponds, growing plants and fish. Funding and technical assistance from GRET-Professionals for Fair Development have also supported community-based activities in cleaning and maintaining the ponds.



Restoration of Wetlands for Green Spaces
Objective: To enhance the quality of the life of citizens through, improved green spaces, ecotourism and ecological diversity, new tourism enterprises for those living around the wetlands, wastewater treatment and storm water retention to prevent floods, and restoration of heritage values of the city.

Given the funding and sustainability issues of the current effort, deliberation is on-going to incentivise people to take larger role in preserving the wetlands.

Vientiane, Lao PDR



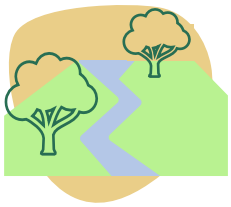
Drainage System and Environmental Improvement
Objective: To effectively improve the operation and maintenance of the drainage system to reduce flooding incidents.

Management of drainage and waste have been a key aspect of Vientiane's priority in mitigating flood and increasing water reserves.

Korea Environmental and Industrial Technology Institute (KEITI) supported the city in undertaking a pre-feasibility study on the drainage system and environmental improvement. Vientiane seeks to collaborate with partners on both financial and technical assistance to advance this effort into the construction phase.

Johor Bahru, Malaysia

Skudai River plays a pivotal role in Johor Bahru's effort in managing its water resources. It provides raw water for the Skudai Water Treatment Plant, facilitates fishery activities, and can potentially be revived as a mode of transport. In this regard, Iskandar Regional Development Authority (IRDA) has been implementing river rejuvenation programme focusing on three key strategies, namely, to boost socio-economic value, sustain environment and bio-diversity, and leverage on technology.



Management of Water Resources and Distribution
Objective: To upgrade existing water plants with advanced technology and smart water management systems towards increasing water reserves.

Leveraging on technology, IRDA has developed the River Management and Monitoring Tool (RMMT) to consolidate data, analyse issues, and formulate necessary actions and monitoring for the rivers. The RMMT dashboard and analytics have also been integrated to the Iskandar Malaysia Urban Observatory (a centralised big data platform) to allow the different agencies to effectively plan and execute inter-agency action plans and programmes.

After completing the feasibility study in 2022 on applying sensor, automation, and IoT solution to the RMMT, work is underway to procure for the Proof of Concept for data sensor automation and monitoring system at Skudai River. Besides the challenge to secure funding for IoT installation, Johor Bahru also needs to ensure the local agencies readiness to adopt and adapt to digitalisation, as well as, continuous integration of the current RMMT system with those of other related agencies.

Kuching, Malaysia

Sarawak Multimedia Authority (SMA) has embarked on developing a comprehensive and flood-related database utilising real-time data collection which aims to enable a more accurate flood modelling and prediction.

In Sarawak, 391 out of 500 telemetry station and system have been upgraded of which 78 are located in Kuching. Four Intelligent Gauges have been implemented to detect flash flood (water level detection at road level). Data and alerts are provided to the citizens via IHYDRO, an online Hydrological Telemetry System developed to provide a real-time flood and drought monitoring.

Integrated Flood Management and Response System
Objective: To create a comprehensive and real-time flood-related database through the upgrading critical rainfall/ water level telemetry-enabled stations for more accurate flood modelling and prediction to prevent and mitigate floods.

50 new critical rainfall/water level telemetry-enabled stations and Intelligent Gauges will be installed or upgraded from this year until 2025. Availability of funding and innovative solutions remain a challenge to deliver the desired outcomes. SMA has been in partnership with JTEC and MLIT Japan (under SmartJAMP) in research, capacity building, as well as transfer of knowledge, experience, and expertise in the areas of flood management and response.

Built Infrastructure

The focus area of Built Infrastructure advocates investment in infrastructure to deliver multiple benefits across various stakeholders, whether private or public. These can include investing in utilities such as energy, mobility and transportation, as well as buildings and construction.

Built Infrastructure accounts for 23% of the overall ASCN projects. Phnom Penh, Kota Kinabalu, Davao City, Bangkok, and Chonburi are examples of ASCN Cities that have projects in this focus area.

Phnom Penh, Cambodia

The City Bus Authority (CBA) of Phnom Penh has continued the efforts to enhance the efficiency of Phnom Penh public transport through increasing the number bus routes, units, and facilities; improving bus operation; providing mobile application for real-time traffic tracking and cashless payment system.

Exhibit 11: Improving Efficiency of Public Transport
Objective: To improve the efficiency of public bus service.

COVID-19 had resulted in a massive drop in the number of passengers which stood at 1.6 million in 2022, compared to 8.9 million in 2019. As of April 2023, 4 out of 13 bus routes and 60 out of 235 buses have been in operation. CBA envisage all 13 bus routes and 235 buses to be fully in operation by end of this year.

Under the Project for Improvement Bus Operation (PIBO) spanning from 2017-2022, JICA had supported CBA in developing and operating the bus routes including through the development of Bus Operation Management System which is part of the Traffic Control Centre located in the Phnom Penh City Hall. PIBO also focused on improving maintenance and inspection capacity, conductors and drivers management capacity, business management capacity, and policy planning for public transport priority measures. To advance the current efforts, provision of capacity building for public transport staff, enhancing traffic control system, and using innovative financial tools to invest in mass transit are necessary.

Kota Kinabalu, Malaysia

Guided by its Integrated Public Transportation Masterplan, Kota Kinabalu works towards achieving a well-planned, efficient and systematic transportation system and promotes public transportation as an everyday viable mobility option. The Masterplan focuses on enhancing some key aspects, namely, convenience, information, proximity, destination choices, directness, and security.

The revamp of Central Integrated Bus Terminal was completed and will be followed by the construction of integrated bus terminals in the northern and southern parts of the city to be implemented through joint venture model. Deliberation is on-going on the development of bus rapid transport and light rail transport. Lack of funding for and technical expertise in public transport system, route alignment, and lack of ridership are some of the barriers to a successful integrated public transport system. Strong support at the state and federal levels, as well as, an inclusive mechanism for the planning, design, and implementation that incorporates input of residents, businesses, and end users are of paramount importance.

Exhibit 12: Design of the Northern Inanam Integrated Bus Terminal

Integrated Public Transport System
Objective: To develop an efficient and integrated public transport system for Kota Kinabalu.

The Northern Inanam Integrated Bus Terminal plans to provide **174** buses, **1374** industrial units, and **18** industrial parks.

Davao City, the Philippines

Formerly known as High Priority Bus System Project, Davao Public Transport Modernisation Project (DPTMP) includes public transport system improvement, institutional strengthening, and social development program. ADB has been supporting Davao City Government in this undertaking. Forming part of DPTMP is the development of the High Priority Bus Information System (HPBIS) which has been completed to serve as a web-based application for the Social Development Program (SDP) for the Project Affected Persons of DPTMP on Data Banking (HPBIS-SDP), Grievance Management Operations (HPBIS-Grievance System), and SMS blast services.

Intelligent Transportation and Traffic Systems with Security
Objective: To enhance transportation and traffic management capabilities by leveraging on the latest technological innovations available and ensuring inclusivity and security.

Complementing DPTMP, Davao City Government has been working on the implementation of Davao Bus Driver Training Technology Pilot, Procurement & Framework Project, supported under the AASCTF.

Bangkok, Thailand

Since November 2021, Krung Thep Aphiwat Central Terminal Station or Bang Sue Grand Station has been operating metro rail system. Next endeavour is to construct high-speed train service to the north eastern part of Thailand (first section) and three Airports High-Speed Rail Link.

Bang Sue Grand Station has been designed using Transit Oriented Development (TOD) approach and Smart City concept and is forming part of the larger Bang Sue Smart City Development Plan. Construction of government office area and smart business complex are two leading projects of the Bang Sue Smart City which also will be using TOD approach.

Exhibit 13: Bang Sue Grand Station
Objective: To function the Bang Sue Central Railway as the hub for the new Commuter Rail System that provides new lines serving the areas near the city centre, while serving as the main stop for the high-speed rail; as well as to overcome the existing infrastructural pressure on the existing terminal while simultaneously reducing commuting time.

Chonburi, Thailand

In partnership with AMATA Corporation PCL, an industrial city developer, Chonburi has been implementing the Smart Grid that focuses on power quality, price, and response time.

Since 2020, the Supervision Control and Data Acquisition (SCADA) has been used to support the operation of the power plant, distribution network, and load management. In 2021, the Meter Data Management, Smart Meters, and Customer Relation Management (CRM) were completely installed, intelligent solar rooftops and was developed. The Meter Data has been expanded to interface with the Advanced Distribution Management System (ADMS). The ADMS will be completed and rolled out in the fourth quarter of 2023.

Challenges to the implementation relate to the protocol to communicate to the various Intelligent Electronic Devices (IEDs) and systems, the optimum cost to install new IED to get the highest reliability and stability, as well as the high cost of Energy Solar System (ESS) for commercial use.

Exhibit 14: Smart Data Utilisation for Smart Microgrid Development
Objective: To improve the management of electrical networks, generation systems, transmission systems and power distribution systems, with a systematic energy management and energy storage structure.

Industry & Innovation

The focus area of Industry and Innovation pertains to encouraging industries and businesses to capitalise on new technologies, using innovation as a catalyst to build competitive advantages and transform processes. Industry and Innovation accounts for 8% of the overall ASCN projects. Bandar Seri Begawan, Banyuwangi, Jakarta, and Singapore are examples of ASCN Cities that have projects in this focus area.

Bandar Seri Begawan, Brunei Darussalam

Through the Digital Payment Hub, digital technologies will be utilised to enable commercial and socio-economic benefits for the citizens by connecting the unbanked, ensuring instant fund transfers, reducing transaction costs, stimulating financial sector innovation, improving ecosystem stability, and improving transaction confidence. Pricing scheme, participants' readiness, risk management, and regulatory environments are the elements that are being addressed to facilitate the implementation of the Digital Payment Hub. The National Digital Payments Network Sdn Bhd has been appointed to implement the Digital Payment Hub.

Digital Payment Hub

Objective: To enable account-to-account fund transfers that are secure, instant and cost-effective on a 24x7x365 basis as well as introduction of instant payment system across local banks, standard national QR code, and central addressing scheme local payment gateway.

Banyuwangi, Indonesia

Collaboration with the private sector is key to Banyuwangi Regency Government's efforts to empower youth. Through its programme called *Jagoan Banyuwangi*, or Banyuwangi Champs, Banyuwangi Regency Government has partnered with Jagoan Indonesia, a business incubator based in East Java, Indonesia, that focuses on connecting small and medium-sized enterprises (SMEs) to modern retail outlets. Jagoan Banyuwangi has three sub-programmes that direct support on the areas of agriculture, entrepreneurship, and digital respectively.

Jagoan Tani is a competition to advance the role of agriculture in Banyuwangi's economic development that puts youth in the driving seat. Winners are awarded a grant to grow their agri-businesses. Meanwhile, *Jagoan Bisnis* is a training suite designed to help young entrepreneurs to spearhead local products in non-agriculture sectors. Submitted ideas are included in an incubator programme with experienced mentors. Winners are awarded seed funding to kick-off their business ideas. On the other hand, *Jagoan Digital* is an incubator programme dedicated to provide support and resources for digital start-ups. The programme includes mentorship, bootcamp, and pitch desk. The three sub-programmes have been implemented from 2021-2022 and new batches will be implemented later this year.

Spearing Industrial Growth through Education

Objective: To collaborate with private entities to provide the youth with knowledge in e-commerce and online trading through the inclusion of customised IT modules in the skills curriculum.

Exhibit 15: Achievements of Jagoan Banyuwangi

Jakarta, Indonesia

Through a platform called JakPreneur, Jakarta Provincial Government has supported MSMEs in various aspects such as basic and advanced level trainings, licensing, marketing, financial reporting, as well as capital facilities. Partnerships with various tech companies, financial institutions, and education institutions have also been intensified to create an entrepreneurial ecosystem for the MSMEs and sustain it.

Since the launch of JakPreneur in February 2020 until June 2023, 356,298 MSMEs from various business sectors have registered with JakPreneur. A total of 5,580 culinary-based SMEs assisted by JakPreneur have received free halal certification. Around 79 JakPreneur Bazaars promoting MSMEs from various industries have already been held in 2023.

JakPreneur
Objective: To encourage economic growth through supporting the MSMEs in Jakarta for long-term cooperation by providing training and mentoring to gain capital which could be accessed through digital platforms.

To advance the current implementation, the development of an e-marketplace ecosystem through JakPreneur is necessary to facilitate digital payment, online transaction training, and product marketing. In this regard, continued collaboration between Jakarta Provincial Government, Financial Institutions, and startups will be key to ensure the sustainability of MSMSEs' businesses and enable expansion of their market size beyond Jakarta towards increasing the economic growth.

Singapore

Singapore strives to digitalise industries to increase business efficacy and create new jobs and opportunities. Greater convenience and accessibility are being provided to citizens and businesses through, among others, PayNow, GoBusiness, and SGQR.

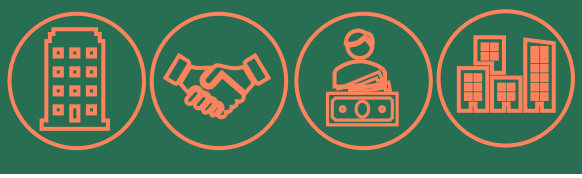
PayNow is a peer-to-peer funds transfer service available to retail customers of participating banks and participating Non-Bank Financial Institutions (NFIs) in Singapore. PayNow allows sending and receiving money instantly via mobile number or virtual payment access proxy. The service has also broadened to facilitating fund transfers & remittances between Singapore and Thailand as well as India. In 2022, PayNow enabled 311 million individual and corporate transactions.

GoBusiness platform continues to connect businesses to over 300 of government e-services and resources.

SGQR is a single standardised QR code for e-payments and combines multiple payment schemes into a single SGQR label, making QR payments simple for both consumers and merchants. Consumers just need to look out for the SGQR label to see which payment options a merchant accepts, while merchants only need to display one SGQR label instead of many labels. The SGQR scheme is co-owned by the Monetary Authority of Singapore and InfoComm Media Authority of Singapore.

Efforts have also been invested to build stronger defences to safeguard people and improve the resilience of government ICT infrastructure against cyber-attacks. Since 2022, 200,000 calls were blocked and over 3.5 million scam messages were detected. Cloud migration has been done for 64% of eligible government services and it is expected to reach 70% in 2023. AI and data analytics are also being used to defend government systems, as well as strengthen threat detection, and incident response.

Smart Nation
Objective: To transform Singapore using technology in the four pillars of digital government, digital economy, digital society, and smart cities.



3| Observations and Recommendations

The varying capacities and journeys of cities in implementing the ASCN projects provide opportunities for greater collaboration amongst ASEAN and with its partners. The ASCN Discussion Series initiated in 2023 leverages on these differences, and connects cities and partners through exchange of knowledge, experiences, and practices to promote smart solutions and sustainable actions.

Every digital transformation requires investment in digital infrastructure, enabling data governance and regulatory framework, as well as investment in digital literacy for the human resource. Harnessing innovative digital technologies and solutions is essential for advancing smart city development. However, smart city projects are oftentimes considered as expense instead of investment. Therefore, local governments need to be able to provide strong economic justification or business case for smart city projects to attract funding and financing. Identifying priorities and needs of national and local governments, utilising the approaches that are relevant for their respective contexts, and taking practical actions will be important. In this regard, the ASEAN Smart City Investment Toolkit that is currently being designed requires a holistic approach and cooperation from various stakeholders so that once developed, the toolkit can support their respective smart city journey.

Ensuring the availability of digital infrastructure must come hand in hand with efforts in building digital capacity within the local governments to make use of data to help improve urban life. Collaborating with the Seoul National University supported by the ASEAN-Republic of Korea Cooperation Fund, the ASCN has co-curated capacity building activities on data-driven smart city planning. Through the ASEAN Smart City Professional Program, short-term training, master's degree course, and knowledge platform will be implemented from 2023 to 2027 to build digital talent and support national and local government in data-driven smart city planning.

Nurturing a culture of sharing and building trust among agencies will be key to the provision of public services. By enabling all stakeholders to work together with a holistic view, ASCN cities can promote collaboration with other cities, the private sector, civil society, academia and external partners to improve the quality of life for the peoples, while increases the efficiency of resource allocation.

Appendix

List of ASCN Projects across the Six Focus Areas

No.	Country	City	Project	Focus Area
1.	Brunei Darussalam	Bandar Seri Begawan	Revitalisation of Kampong Ayer (Water Village)	Civic & Social
2.		Bandar Seri Begawan	Clean River Management	Quality Environment
3.		Bandar Seri Begawan	Digital Identity	Civic & Social
4.		Bandar Seri Begawan	Digital Payment Hub	Industry & Innovation
5.		Bandar Seri Begawan	National Information Hub	Civic & Social
6.		Bandar Seri Begawan	Gov.bn	Civic & Social
7.		Bandar Seri Begawan	Bus Passenger Information System	Built Infrastructure
8.	Cambodia	Battambang	Urban Street and Public Space Management	Built Infrastructure
9.		Battambang	Solid and Liquid Waste Management	Quality Environment
10.		Phnom Penh	11 Sidewalks Rejuvenation	Built Infrastructure
11.		Phnom Penh	Improving Efficiency of Public Transport	Built Infrastructure
12.		Phnom Penh	Smart City Strategic Planning	Civic & Social
13.		Phnom Penh	Phnom Penh Smart City Hub	Industry & Innovation
14.		Siem Reap	Smart Tourist Management System	Civic & Social
15.		Siem Reap	Solid Waste and Wastewater Management	Quality Environment
16.	Indonesia	Banyuwangi	Spearing Industrial Growth through Education	Industry & Innovation
17.		Banyuwangi	Creating Inclusive Economic Growth through Tourism-Based Development	Civic & Social
18.		Banyuwangi	Smart Kampung	Civic & Social
19.		Banyuwangi	Caring for Stunting	Health & Well-Being
20.		Banyuwangi	Sustainable Waste Management	Quality Environment
21.		Jakarta	JakPreneur	Industry & Innovation
22.		Jakarta	JakLingko	Built Infrastructure
23.		Jakarta	JAKI	Civic & Social
24.		Jakarta	Jakarta Smart City Goes to School	Civic & Social

No.	Country	City	Project	Focus Area
25.		Makassar	Smart Health Care	Health & Well-Being
26.		Makassar	Smart E-Tax Services	Civic & Social
27.		Makassar	Tourism Alleys	Civic & Social
28.		Makassar	Makassar Virtual Economic Centre	Industry & Innovation
29.		Makassar	Makassar Incubator Centre	Industry & Innovation
30.	Lao PDR	Luang Prabang	Restoration of Wetlands for Green Spaces	Quality Environment
31.		Luang Prabang	Construction of Concrete Roads for Communities	Built Infrastructure
32.		Luang Prabang	Smart City Planning and Development	Civic & Social
33.		Luang Prabang	Smart and Integrated Urban Strategy	Civic & Social
34.		Vientiane	Drainage System and Environmental Improvement	Quality Environment
35.		Vientiane	Vientiane Sustainable Urban Transport	Built Infrastructure
36.	Malaysia	Johor Bahru	Iskandar Malaysia Urban Observatory	Civic & Social
37.		Johor Bahru	Management of Water Resources and Distribution	Quality Environment
38.		Johor Bahru	Smart Integrated Mobility Management System	Built Infrastructure
39.		Johor Bahru	Iskandar Malaysia Integrated Urban Services Program	Safety & Security
40.		Kota Kinabalu	Integrated Public Transport System	Built Infrastructure
41.		Kota Kinabalu	Integrated Solid Waste Management	Quality Environment
42.		Kota Kinabalu	Smart New Township and Smart Water Management	Quality Environment
43.		Kuala Lumpur	Kuala Lumpur Urban Observatory/G-Asset	Civic & Social
44.		Kuala Lumpur	Kuala Lumpur Integrated Submission System	Civic & Social
45.		Kuala Lumpur	GoKL Journey Planner	Built Infrastructure
46.		Kuala Lumpur	Smart Bin	Quality Environment
47.		Kuala Lumpur	Bicycle Friendly City	Quality Environment
48.		Kuching	Integrated Smart Traffic Light System	Built Infrastructure
49.		Kuching	Integrated Flood Management and Response System	Quality Environment
50.		Kuching	Introduction of Blockchain Technology	Civic & Social
51.	Myanmar	Nay Pyi Taw	Affordable and Low-Cost Housing	Health & Well-Being
52.		Nay Pyi Taw	Comprehensive University	Health & Well-Being
53.		Nay Pyi Taw	Smart Street Lighting	Safety & Security

No.	Country	City	Project	Focus Area
54.		Nay Pyi Taw	Safe City	Safety & Security
55.		Mandalay	Traffic Congestion Management	Built Infrastructure
56.		Mandalay	Solid Waste and Water Waste Treatment	Quality Environment
57.		Mandalay	On-grid Solar System at the Water Pumping Stations	Quality Environment
58.		Mandalay	Cadastral Map and GIS Database	Civic & Social
59.		Mandalay	Yandanabon Park	Quality Environment
60.		Yangon	Conservation of Downtown Yangon	Civic & Social
61.		Yangon	Transit Oriented Development in Hlaing Thar Yar Township	Built Infrastructure
62.		Yangon	One Map Yangon	Civic & Social
63.	Philippines	Cebu City	Cebu City Bus Rapid Transit	Built Infrastructure
64.		Cebu City	Cebu City Automated Citywide Traffic System	Safety & Security
65.		Davao City	Converged Command and Control Center	Safety & Security
66.		Davao City	Intelligent Transportation and Traffic Systems with Security	Built Infrastructure
67.		Manila	Command Center Upgrade	Safety & Security
68.		Manila	E-Government Services	Civic & Social
69.	Singapore	Singapore	E-Payments	Industry & Innovation
70.		Singapore	National Digital Identity	Civic & Social
71.		Singapore	Smart Nation	Industry & Innovation
72.	Thailand	Bangkok	Bang Sue Smart City	Quality Environment
73.		Bangkok	Bang Sue Grand Station	Built Infrastructure
74.		Chonburi	Waste to Energy	Quality Environment
75.		Chonburi	Smart Data Utilisation for Smart Microgrid Development	Built Infrastructure
76.		Phuket	Artificial Intelligence-Led Image Traffic Visualisation and Analysis	Built Infrastructure
77.		Phuket	Mobility-as-a-Service	Built Infrastructure
78.		Phuket	Smart Pier	Safety & Security
79.		Phuket	City Data Platform	Civic & Social
80.		Phuket	Phuket Health Sandbox	Health & Well-Being
81.	Viet Nam	Da Nang	Intelligent Operation Centre	Safety & Security
82.		Da Nang	Smart Water Management System	Quality Environment

No.	Country	City	Project	Focus Area
83.		Ha Noi	Intelligent Operation Centre	Safety & Security
84.		Ha Noi	Transport Operation and Surveillance Centre	Built Infrastructure
85.		Ho Chi Minh City	Integrated Operation Centre	Civic & Social
86.		Ho Chi Minh City	Integrated and Unified Emergency Response Centre	Safety & Security